

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	<i>Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.</i>		
Student Name:	GAUTHAM G	Reg. No.:	192524371
Section / Batch:	SLOT-B	Date:	28/08/2026

Code of Conduct

I am GAUTHAM G (192524371) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student : GAUTHAM G

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:		100 Marks	

Submission Evidence — Answer Screenshots

Assessment Completion Summary

All 5 questions answered and saved (5/5)



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My Assessments

Course: CSA0309RB

PSEUDOCODE WRITING TASK

OPEN

✓ Completed

CO5_AT1_PSEUDOCODE WRITING TASK

5/5 answered

CO3_AT3_MCQ

OPEN

✓ Completed

Multiple Choice Questions

25/25 answered

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

OPEN

✓ Completed

Debugging & Optimization Task

10/10 answered

Question 1 — Minimum Spanning Tree (Kruskal's Algorithm)

Saved Answer Screenshot — 20 Marks

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Q1
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

Your Answer

4. For each edge (u, v) in sorted edges:
If u and v are in different sets:
Add (u, v) to MST
Join the sets of u and v

If MST has $V - 1$ edges:
Stop

Save Answer

Next →

5/5 answered

Question 2 — Graph Sorting (Topological Sort)

Saved Answer Screenshot — 20 Marks

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Q1

Minimum Spanning Tree -
Kruskal's Algorithm

20 marks

Q2

Graph - Topological Sort

20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm

20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm

20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm

20 marks

5/5 answered

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- a. Detect cycle before sorting
- b. Use DFS approach

Your Answer

3. Mark v as VISITED

4. Push v into stack

Simple Explanation

Use DFS to visit each task.

Save Answer

Next →

Question 3 — Shortest Path Algorithm (MST & Dijkstra's Algorithm)

Saved Answer Screenshot — 20 Marks

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Q1
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- Use MST for infrastructure
- Use shortest path for routing

Your Answer

```
SmartTransportation(G, Source)
```

- MST = Kruskal(G)
- Build transportation network using MST
- Set distance of all vertices = INFINITY

Save Answer

Next →

Question 4 — Shortest Path Algorithm (Dijkstra's Algorithm)

Saved Answer Screenshot — 20 Marks

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QUESTIONS

Q1
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

DijkstraWithVoucher(G, S, D)

1. Set all distances = INFINITY
2. distance[S][0] = 0
3. distance[S][1] = 0
// 0 = voucher unused, 1 = voucher used

Save Answer

Next →

Question 5 — Minimum Spanning Tree (Prim's or Kruskal's Algorithm)

Saved Answer Screenshot — 20 Marks

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QUESTIONS

Q1
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm 20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

SecondBestMST(G)

1. Find MST using Kruskal's Algorithm
2. Store MST and its total cost
3. secondBest = INFINITY

Save Answer

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____